INTERACTIVE CREDIT RISK XGBOOST DEBUGGER USING EXPLAINABLE ARTIFICIAL INTELLIGENCE FOR LOAN APPROVAL PREDICTION

ABSTRACT

Financial institutions increasingly use machine learning models to automate loan approval and credit risk assessment. Among various algorithms, XGBoost is widely used because of its high predictive performance and efficiency. However, despite its accuracy, XGBoost often behaves like a black-box model, making it difficult for users to understand why a loan application is approved or rejected. This lack of transparency reduces trust, complicates debugging, and may introduce bias in decision-making systems.

This project proposes an **Interactive Credit Risk XGBoost Debugger using SHAP (SHapley Additive Explanations)** and Streamlit to improve model interpretability and transparency. The system allows users to upload a credit risk dataset, train an XGBoost classification model, evaluate prediction performance, and visualize feature contributions using SHAP explainability techniques.

The application provides interactive dashboards for feature importance analysis, prediction explanation, and model debugging. Users can understand which features positively or negatively influence loan approval decisions. The project aims to make machine learning systems more explainable, trustworthy, and user-friendly for data science students, researchers, and financial analytics applications.

CHAPTER 1: INTRODUCTION

1.1 INTRODUCTION

Machine learning is increasingly used in financial sectors for tasks such as fraud detection, risk assessment, and loan approval prediction. Financial institutions process large volumes of loan applications, making manual assessment time-consuming and prone to human bias.

Loan approval prediction systems help institutions automate decision-making by learning patterns from historical data. However, many machine learning models act as **black-box systems**, meaning they provide predictions without explaining why decisions are made.

This project addresses this issue by developing an explainable credit risk prediction system that combines **XGBoost**, **SHAP Explainable AI**, and an interactive **Streamlit dashboard**.

The system not only predicts loan approval outcomes but also explains:

- Why a loan was approved
- Why a loan was rejected
- Which features influenced the decision
- Overall model health and reliability

1.2 PROBLEM STATEMENT

Traditional loan approval systems face several limitations:

- Manual processing is time-consuming
- Decisions may be subjective
- Complex machine learning models lack transparency
- Users cannot understand prediction reasons

- Model performance issues such as overfitting and class imbalance are often ignored

There is a need for a system that provides:

- Accurate loan prediction
- Transparent decision explanations
- Model debugging and validation

1.3 PROPOSED SOLUTION

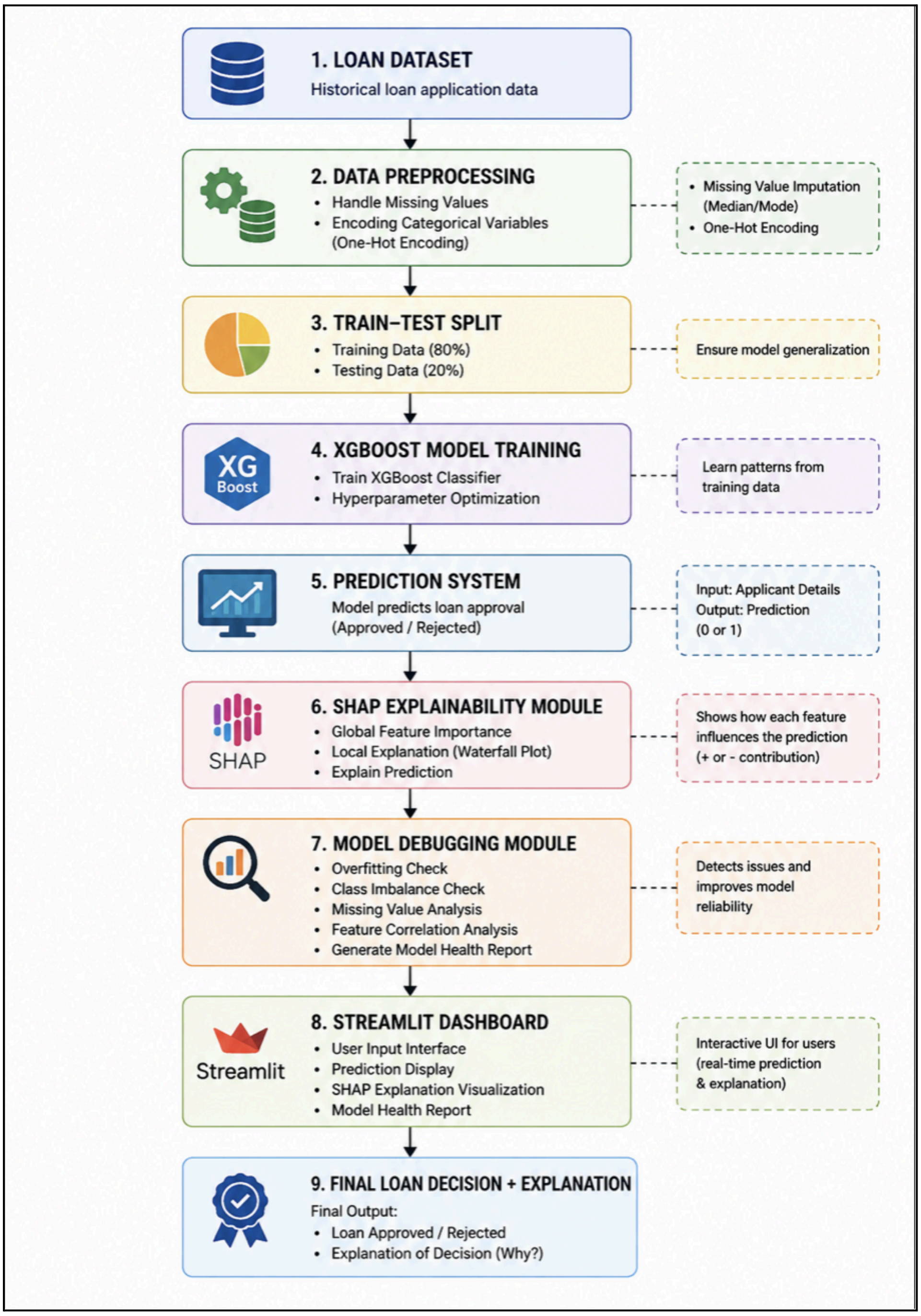
The proposed system develops an **Interactive Credit Risk XGBoost Debugger using Explainable AI**.

The system performs the following tasks:

- Preprocesses loan application data
- Trains an optimized XGBoost model
- Predicts loan approval outcomes
- Uses SHAP to explain prediction decisions
- Detects model health issues
- Displays interactive visualizations through Streamlit

The solution combines prediction accuracy with interpretability and debugging functionality.

CHAPTER 2: SYSTEM ARCHITECTURE



The proposed system architecture describes the complete workflow of the **Interactive Credit Risk XGBoost Debugger using Explainable Artificial Intelligence (XAI)**. The system processes loan application data, trains a machine learning model, generates predictions, explains the model decisions, and provides interactive visualization through a dashboard.

Step 1: LOAD DATASET

The process begins with collecting the loan dataset containing historical applicant information. The dataset includes attributes such as:

- Applicant income
- Co-applicant income
- Loan amount
- Credit history
- Education
- Marital status
- Property area
- Dependents

These historical records act as input data for model training and prediction.

Step 2: DATA PREPROCESSING

Raw datasets often contain missing values and categorical variables that cannot be directly processed by machine learning algorithms. Therefore, preprocessing is performed to improve data quality.

The preprocessing stage includes:

- Handling missing values
    - Median for numerical variables
    - Mode for categorical variables

- Converting categorical variables into numerical format using:

One-Hot Encoding

This ensures that the dataset becomes suitable for machine learning operations.

Step 3: TRAIN_TEST SPLIT

The processed dataset is divided into training and testing datasets.

The dataset split used in this project is:

- Training Data = 80%
- Testing Data = 20%

The training dataset is used to train the model, while the testing dataset evaluates performance on unseen data and checks generalization ability.

Step 4: XGBOOST MODEL TRAINING

The training dataset is provided to the **XGBoost classifier**.

The model learns relationships between input features and loan approval outcomes.

Hyperparameter optimization techniques are used to improve:

- Accuracy
- Generalization capability
- Overfitting reduction

Step 5: PREDICTION SYSTEM

Once training is completed, the model predicts whether a loan application should be:

- Approved (1)
- Rejected (0)

The prediction system accepts user-provided applicant details and generates the final prediction.

Step 6: SHAP EXPLANABILITY MODULE

Machine learning models often behave as black-box systems. Therefore, SHAP (SHapley Additive Explanations) is integrated to improve interpretability.

SHAP performs:

- Global feature importance analysis
- Individual prediction explanations
- Waterfall visualizations
- Feature contribution analysis

Positive SHAP values increase approval probability, while negative SHAP values reduce approval probability.

Step 7: MODEL DEBUGGING MODULE

To ensure reliability, a debugging module is included.

The module performs:

- Overfitting detection
- Class imbalance checking
- Missing value analysis

- Feature correlation analysis
- Model health report generation

This helps identify issues affecting model performance.

Step 8:STREAMLIT DASHBOARD

A Streamlit-based interactive dashboard is developed to provide a user-friendly interface.

The dashboard allows users to:

- Enter applicant information
- View predictions
- Explore SHAP explanations
- Analyze model health reports

Step 9: FINAL LOAN DECISION AND EXPLANATION

The final output generated by the system includes:

- Loan Approval or Rejection decision
- Approval probability
- Explanation of factors influencing the prediction
- Visualization of important contributing features

Thus, the system not only predicts loan decisions but also explains *why* a particular decision was made, improving transparency and trust.

CHAPTER 3: METHODOLOGY

The project follows these steps:

Step 1: DATA COLLECTION

The loan dataset containing applicant details was loaded.

Features include:

- Applicant income
- Co-applicant income
- Loan amount
- Credit history
- Education
- Property area
- Marital status
- Dependents

Step 2: DATA PREPROCESSING

The preprocessing stage included:

- Removing unnecessary columns
- Handling missing values using:
    - Median for numerical variables
    - Mode for categorical variables
- Applying one-hot encoding

Step 3: DATASET SPLIT

The dataset was divided into:

- Training data = 80%

- Testing data = 20%

using:

test_size=0.2

random_state=42

Step 4: MODEL DEVELOPMENT

The XGBoost classifier was trained using optimized hyperparameters:

- max_depth
- learning_rate
- n_estimators
- subsample
- regularization

Step 5 PERFORMANCE EVALUATION

Performance metrics used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Step 6: EXPLAINABILITY INTEGRATION

SHAP Explainable AI was integrated to:

- Calculate feature importance
- Explain individual predictions

- Generate SHAP summary plots
- Generate waterfall plots

Step 7: MODEL DEBUGGING

Model debugging included:

- Overfitting detection
- Class imbalance detection
- Missing value analysis
- Feature correlation analysis
- Health report generation

Step 8:INTERACTIVE DASHBOARD DEVELOPMENT

A Streamlit application was developed for:

- Real-time prediction
- Visualization
- SHAP explanation
- User interaction

CHAPTER 4: TECHNOLOGIES USED

COMPONENT	TECHNOLOGY
Programming Language	Python
Data Processing	Pandas
Visualization	Matplotlib
Machine Learning	XGBoost
Explainable AI	SHAP
Dashboard	Streamlit
Model Evaluation	Scikit-learn
Development Environment	PyCharm

CHAPTER 5: RESULTS

The optimized XGBoost model produced the following performance:

METRIC	VALUE
Training Accuracy	81.46%
Testing Accuracy	78.86%
Precision	75.96%
Recall	98.75%
F1-score	85.87%

📊 Explainable AI Prediction Examples

Example 1: Approved Applicant

FEATURE	VALUE
Applicant Income	5000
Coapplicant Income	0
Loan Amount	150
Credit History	1
Married	Yes
Property Area	Urban

Prediction Result

Loan Status: APPROVED

Approval Probability: 88.77%

Top Positive Factors (SHAP):

FEATURE	SHAP CONTRIBUTION
Credit History	+0.4253
Married	+0.0958
Applicant Income	+0.0765
Loan Amount	+0.0686

Interpretation:

- Credit history strongly increased approval confidence.
- Applicant income positively influenced financial reliability.
- Marital status slightly increased confidence.
- Overall positive factors outweighed risk factors.

Example 2: Rejected Applicant

Input Details

FEATURE	VALUE
Applicant Income	1500
Coapplicant Income	0
Loan Amount	350
Credit History	0
Married	No
Dependents	3+

Loan Status: REJECTED

Approval Probability: 27.90%

Top Negative Factors (SHAP):

FEATURE	SHAP CONTRIBUTION
Credit History	-2.138
Married	-0.1456
CoApplicant Income	-0.1032
Loan Amount	-0.0779

Interpretation:

- Missing credit history strongly pushed the prediction toward rejection.
- The large requested loan amount reduced approval confidence.
- Lack of CoApplicant income reduced financial reliability.
- Overall negative contributions outweighed positive contributions.

Key Insights:

- Credit history was the strongest feature affecting predictions.
- SHAP improved model transparency by showing feature-level contributions.
- Explainability helped understand not only **what the model predicted** but **why it predicted it**.
- The optimized XGBoost model achieved better generalization and reduced overfitting.

CHAPTER 6: STREAMLIT DASHBOARD

To make the machine learning model more interactive and user-friendly, a Streamlit dashboard was developed. The dashboard allows users to perform real-time loan predictions while also understanding the reasoning behind the model decisions through Explainable AI techniques.

The dashboard integrates XGBoost prediction, SHAP explainability, and model debugging functionalities into a single interface.

Main features include:

- Real-time loan approval prediction
- User input interface for applicant information
- Model performance visualization
- SHAP feature importance analysis
- Individual prediction explanations
- Model health analysis and debugging
- Identification of factors increasing and decreasing approval probability.

Dashboard Overview

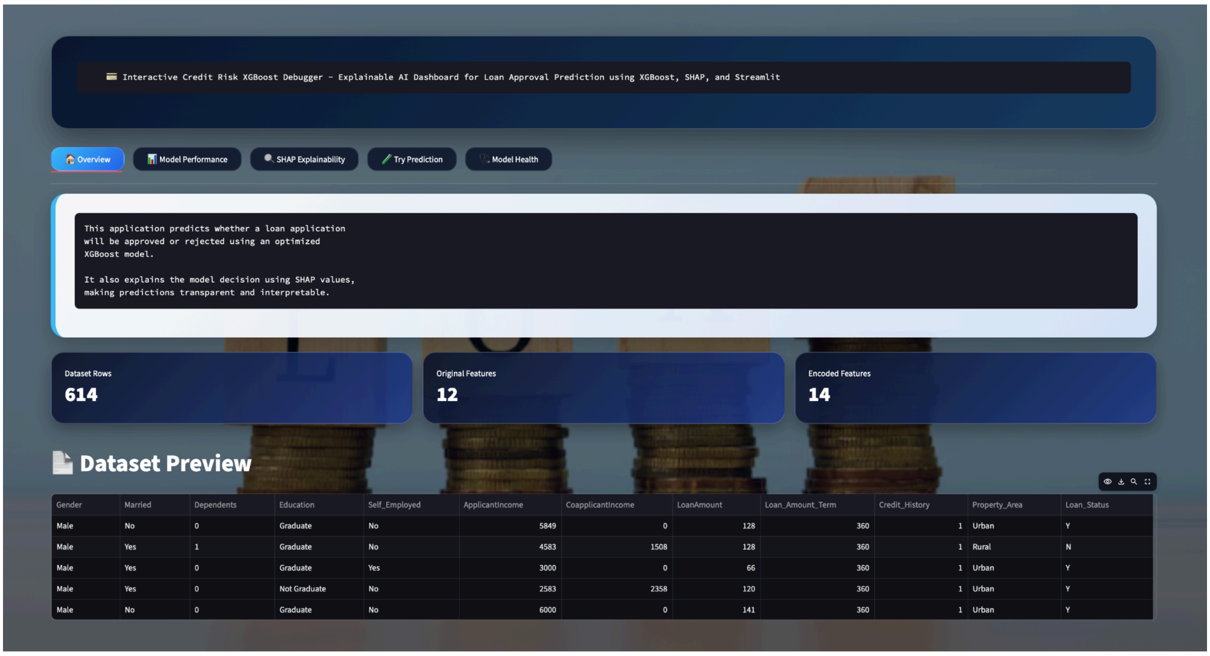


Figure 1: Main dashboard interface showing navigation tabs and system overview.

Description:

The main dashboard provides an overview of the application including project information, dataset details.

Model Performance Dashboard

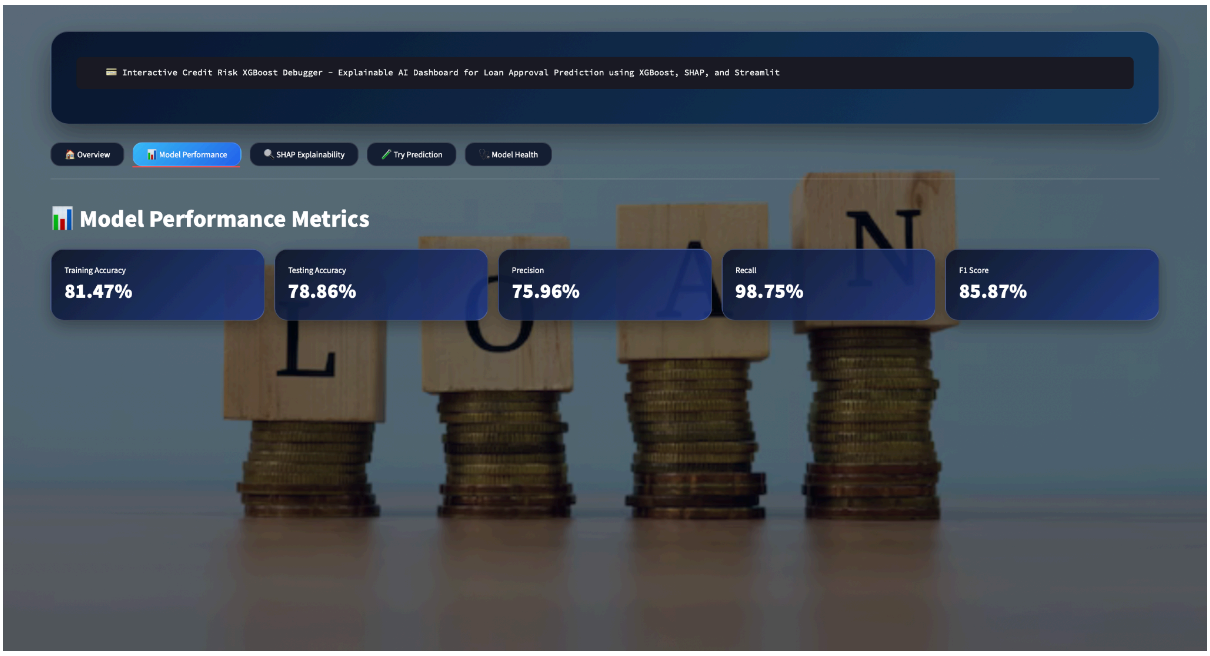


Figure 2: Model performance metrics including training accuracy, testing accuracy, precision, recall, and F1-score.

Description:

The dashboard displays model evaluation metrics to assess the overall predictive performance and generalization capability of the optimized XGBoost model.

SHAP Explainability Dashboard

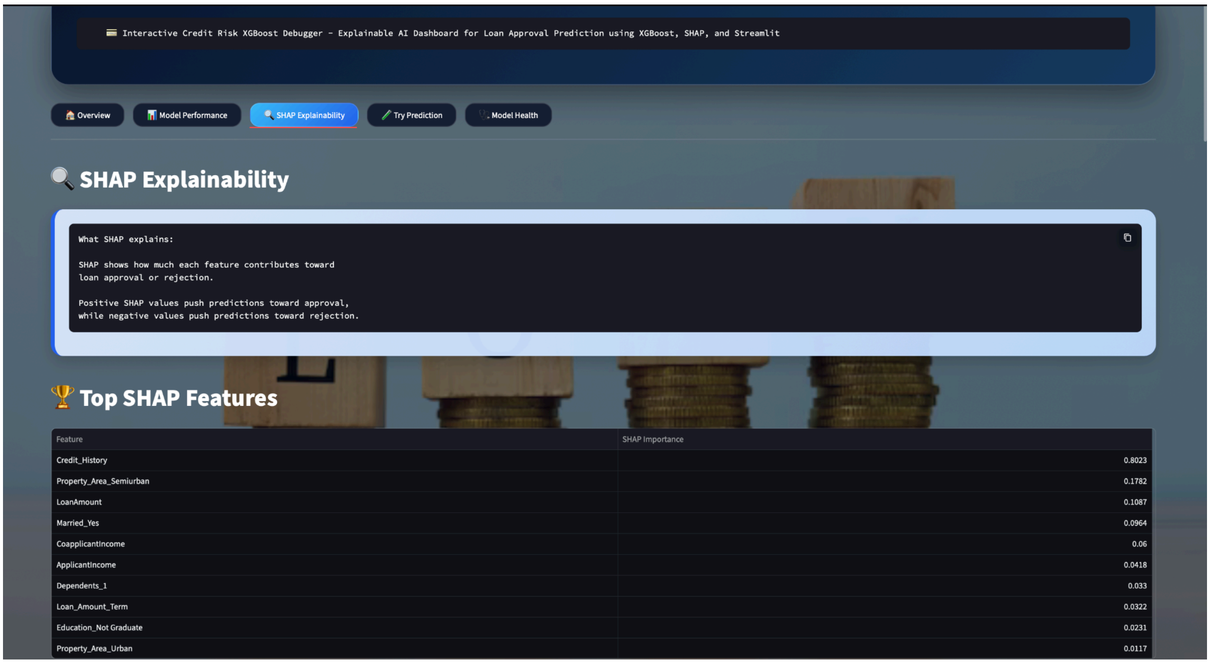


Figure 3: SHAP-based explanation of prediction factors.

Loan Prediction Interface

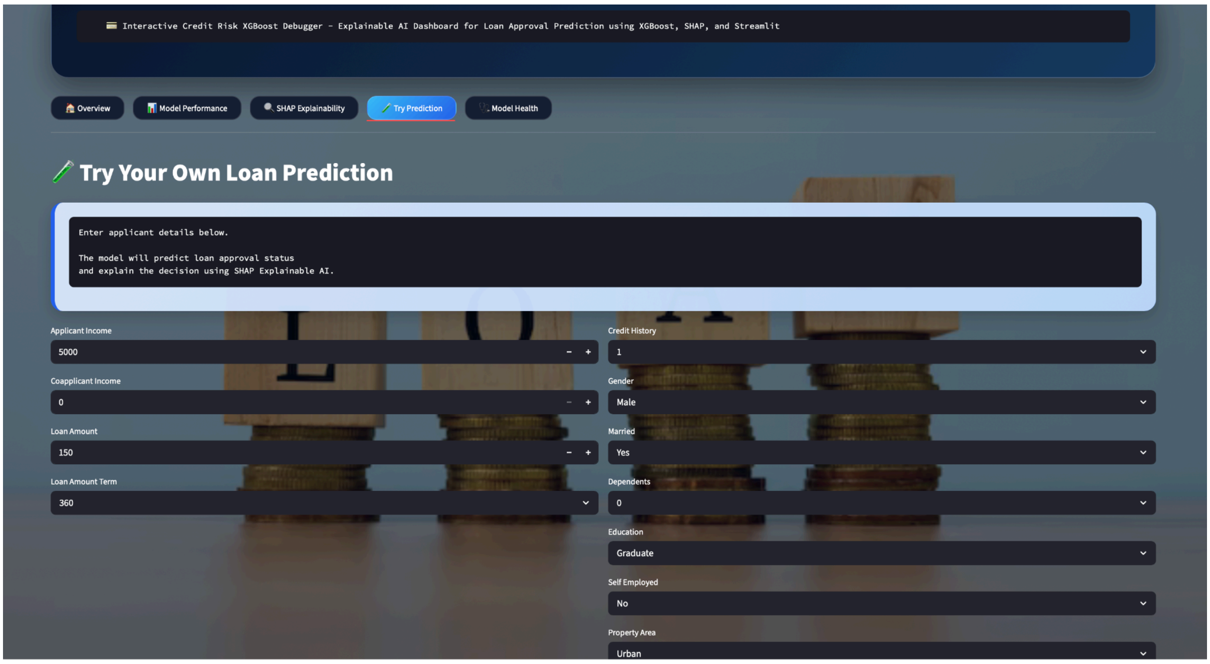


Figure 4: User interface for entering applicant details.

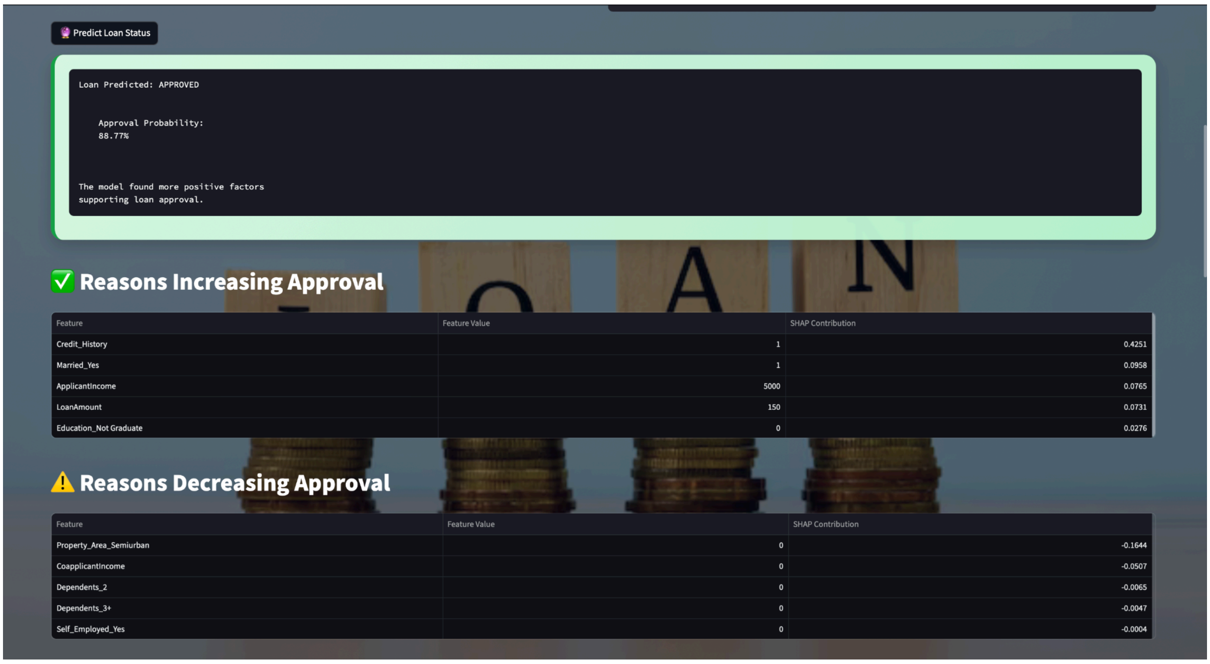


Figure 5: Reasons for Increasing and Decreasing Approval

Figure 6: XAI Decision Summary

Description:

The user can enter loan applicant details such as income, credit history, loan amount, marital status, and other factors. The system processes the information and generates a prediction.

CONCLUSION

This project successfully developed an **Interactive Credit Risk XGBoost Debugger using Explainable AI** capable of predicting loan approval outcomes while providing transparent explanations.

Unlike traditional black-box systems, the developed system integrates SHAP explanations and model debugging techniques to improve trust and reliability. The optimized model achieved improved generalization performance with reduced overfitting.

The developed Streamlit dashboard further enhanced usability by providing an interactive environment for prediction and visualization.

Future improvements may include:

- Deploying the system on cloud platforms
- Using larger datasets
- Comparing additional machine learning models
- Integrating real-time APIs